1. Clinical Question & Core Hypothesis

Clinical Question

Decipher GC predicts metastasis/progression risk but many low-to-intermediate grade tumors show discordant clinical behavior; why do some genomically high-risk tumors not progress while others do? Urine PCA3 and local immune organization (TLS) may modulate or reveal mechanisms of immune containment vs immune evasion that determine progression. 2) Tier 1 paths used: I built on the urinary biomarker (PCA3) and Decipher GC genomic-classifier paths because they provide complementary, clinically actionable readouts—PCA3 as a non-invasive marker of tumor transcriptional activity/shedding and Decipher GC as a tissue-level genomic risk signal. 3) Tier 2 support: Imaging/radiomics provides spatial proxies for immune organization (here operationalized as TLS-like lymphoid aggregates); predictive immune-response markers and in-situ immune infiltrate characterize functional immune competence; metabolite biomarkers reflect tumor metabolic programs that can enable immune escape—these support the mechanism that immune containment (TLS) can modulate genomic risk depending on tumor activity/metabolism (PCA3). 4) Endpoint choice: I selected Progression-Free Interval (PFI) because Decipher is tied to metastasis/progression risk and PFI best captures the clinically relevant progression events we aim to explain. 5) Deliberate exclusions: I excluded weaker, indirect or exploratory links (e.g., SPC/CSF-1 axis, insulin-response, alpha-fetoprotein, therapy-nomogram–Olaparib paths) to keep the hypothesis tightly focused on the PCA3–Decipher–immune imaging interaction that directly addresses progression risk in localized disease?

Working Hypothesis

Primary hypothesis: In localized prostate cancer, the adverse association between a high Decipher GC score and shorter progression-free interval (PFI) is attenuated when TLS-like lymphoid aggregates are present in the tumor and urinary PCA3 levels are low; conversely, absence of TLS-like aggregates or high PCA3 amplify the risk associated with a high Decipher GC score. Secondary hypotheses: radiomics/imaging biomarkers, systemic/metabolic signatures, and in-situ immune infiltrate status will reflect or mediate this interaction and improve non-invasive prediction of PFI. A combined risk model incorporating Decipher GC, TLS status, PCA3, radiomic features, and metabolite markers will outperform single-modality predictors in identifying patients at highest risk for progression and guiding surveillance versus early intervention.

Element	Value
Imaging feature	TLS-like lymphoid aggregates
Anchor gene(s)	PCA3, Decipher GC
Primary endpoint	PFI

2. Why This Gene — Not Others

Primary hypothesis gene(s)

PCA3: PCA3) and Decipher GC genomic-classifier paths because they provide complementary, clinically actionable readouts—PCA3 as a non-invasive marker of tumor transcriptional activity/shedding and Decipher GC as a tissue-level genomic risk signal.

Decipher GC: Decipher GC predicts metastasis/progression risk but many low-to-intermediate grade tumors show discordant clinical behavior; why do some genomically high-risk tumors not progress while others do? Urine PCA3 and local immune organization (TLS) may modulate or reveal mechanisms of immune containment vs immune evasion that determine progression.

Alternative / secondary genes considered

Tier	Gene	Role / Reason excluded from core
Secondary	radiomics/imaging biomarkers	Supporting modifier from knowledge graph
Secondary	predictive markers for immune response	Supporting modifier from knowledge graph
Secondary	metabolites biomarkers	Supporting modifier from knowledge graph
Secondary	status of in-situ immune cell infiltrate	Supporting modifier from knowledge graph

3. Clinically Testable Predictions

Prediction 1: In retrospective cohort data, the prognostic benefit of the imaging feature will be modified by the anchor gene expression level (significant interaction term).

Prediction 2: At the tissue level, tumors with the imaging feature but low anchor gene activity will show immune cells confined to structural peripheries rather than infiltrating tumor nests.

4. Minimum Viable Validation Plan

A. Retrospective Cohort Analysis

Data source: TCGA-BLCA (Firehose Legacy), ~400 patients - Cox PH model with interaction term for the imaging feature and anchor gene - Kaplan-Meier 2x2 stratification

B. Tissue-Level Validation

Methods: Multiplex IF / IHC / spatial transcriptomics

Key readouts: - Immune structure markers (e.g., CD20, CD3, PNAd) - Effector cell localization (CD8) relative to tumor nests - Anchor gene protein / phospho-protein expression - Stromal features (collagen density by SHG, if applicable)

Goal: Confirm that the proposed mechanism operates at the spatial / tissue level.

C. Functional Validation

Models: Ex vivo organotypic cultures, 3D co-culture, syngeneic mouse models

Perturbations: Genetic (KO/rescue) or pharmacologic modulation of anchor gene

Readouts: T-cell infiltration, cytotoxicity (granzyme B / IFN-gamma), tumor growth / survival

Goal: Establish causal relationship between anchor gene activity and immune efficacy.

5. Strengths, Caveats & Alternative Directions

Strengths

- Interaction-first biomarker strategy — asks WHY the imaging feature sometimes fails, not just whether it predicts outcome.

Caveats & Risks

- Hypothesis is computationally derived and requires experimental validation.

Alternative Directions

- radiomics/imaging biomarkers: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.
- predictive markers for immune response: Could serve as an alternative mechanistic axis if the primary hypothesis is not supported.